Easy (10 MCQs)

1.

What minimum clearance do ASHRAE guidelines recommend under raised floors?
A) 6”
B) 18–24"
C) 3–4 ft
D) 1 ft
Answer: B
Recommended plenum height is 18–24" for cooling efficiency ashrae.org+3cleanroom-industries.com+3akcp.com+3123.net+3en.wikipedia.org+3cedengineering.com+3cedengineering.com+4ibm.com+4techtarget.com+4.

2.

What typical power draw per rack defines high-density deployments?
A) 1–2 kW
B) 3–4 kW
C) 10–15 kW
D) 20–30 kW
Answer: C
High-density racks may consume 8–10 kW or more ibm.com+3cleanroom-industries.com+3netzero-events.com+3.

3.

Raised floors are primarily used for:
A) Aesthetics
B) Underfloor cooling and cable routing
C) Sound insulation
D) Fire safety
Answer: B
They support cable routing and underfloor air distribution techtarget.comen.wikipedia.org.

4.

Room ceiling height in modern data centers typically is:
A) 6–7 ft
B) 8–9 ft
C) 10–12 ft
D) Over 15 ft
Answer: B
Standard ceiling heights are 8–9 ft cedengineering.com+15ibm.com+15cleanroom-industries.com+15en.wikipedia.org+1netzero-events.com+1ashrae.org+1en.wikipedia.org+1.

5.

Which factor is not vital for site selection?
A) Nearby utilities
B) Regional tax code
C) Natural disaster risk
D) Network infrastructure
Answer: B
Tax code is less critical than utilities, hazards, connectivity cc-techgroup.com+1lightboxre.com+1cc-techgroup.com+2mullerec.com+2lightboxre.com+2en.wikipedia.org+5theaccessgroup.com+5en.wikipedia.org+5areadevelopment.com.

6.

ASHRAE recommends data center temperatures between:
A) 0–5 °C
B) 8–27 °C
C) 30–40 °C
D) 50–60 °C
Answer: B
ASHRAE acceptable range is 8–27 °C.

7.

Raised floor tiles must support at least:
A) 200 lbs/sq ft
B) 550 lbs point load
C) 50 lbs
D) 1,200 lbs uniformly
Answer: B
Tiles require ≥550 lbs point load capacity mullerec.com+10akcp.com+10en.wikipedia.org+10en.wikipedia.org+15123.net +15en.wikipedia.org+15en.wikipedia.org.

8.

Close-coupled cooling offers:
A) Wider aisles
B) Precision rack-level cooling
C) Raised floor removal
D) VRLA battery backup
Answer: B
Close coupled delivers cooling directly to racks arxiv.org+2arxiv.org+2yourpowerpro.com+2en.wikipedia.orgnetzero-events.com +7en.wikipedia.org+7arxiv.org+7cedengineering.com+1cleanroom-industries.com+1ashrae.org.

9.

Underfloor air distribution uses:
A) Desks
B) Raised plenum for airflow
C) Ceiling vents only
D) Wall fans
Answer: B
UFAD uses underfloor plenums to supply conditioned air en.wikipedia.org+1en.wikipedia.org+1cedengineering.com+2en.wikipedia.org +2en.wikipedia.org+2.

10.

Raised floor helps staging future changes by:
A) Removing tiles
B) Modular remapping
C) Built-in security
D) Cooling oversight
Answer: B
Modularity allows easy reconfiguration en.wikipedia.org123.net.

Medium (20 MCQs)

11.

A rack consuming 100 kW needs how much coolant?
A) No extra
B) 30 kW cooling
C) 10 kW cooling
D) 100 kW cooling
Answer: B
Approx. 30% extra cooling is required ibm.com+2en.wikipedia.org+2ibm.com+2.

12.

Ideal rack footprint with cooling aisles uses:
A) 5 sq ft per rack
B) 30–35% of space
C) 60–70% floor space
D) No planning

Answer: B
IT equipment uses ~30–35% of floor space ashrae.org+2ibm.com+2ibm.com+2.

13.

Raised floor tile loading uniform weight ≈:
A) 1,200 kg/m^2
B) 300 kg/m^2
C) 50 kg/m^2
D) 5,000 kg/m^2
Answer: A
UDL of 12 kN/m^2 ≈1200 kg/m^2 123.net+14en.wikipedia.org+14services.ku.edu+14.

14.

Flood-risk avoidance relates to:
A) Cooling
B) Location near hazards
C) Raised floor height
D) Network speed
Answer: B
Avoiding flood/earthquake zones is essential ibm.com+4lightboxre.com+4cedengineering.com+4cedengineering.com+12mullerec.com+12lightboxre.com+12dataspan.com+6en.wikipedia.org+6ashrae.org+6netzero-events.com.

15.

Immersion cooling reduces energy by:
A) 10%
B) 50%
C) 5%
D) 0%
Answer: B
Immersion can cut energy use ~50% services.ku.edu+15arxiv.org+15en.wikipedia.org+15.

16.

Modular data centers often use:
A) Custom built
B) Shipping-container modules
C) Underground vaults
D) Desktop racks
Answer: B
Modular centers use container-style architecture 123.net+8theaccessgroup.com+8en.wikipedia.org+8.

17.

Raised floor height needed for airflow is:
A) 18–30"
B) 6"
C) 1 m
D) None
Answer: A
Air plenums need 18–30" clearance techtarget.comashrae.org+3cleanroom-industries.com+3ibm.com+3.

18.

HVAC must be sized for:
A) Equipment only
B) Equipment + building + redundancy
C) Building only
D) No oversizing
Answer: B
HVAC sizing accounts for all loads plus overhead 123.net+3services.ku.edu+3techtarget.com+3cedengineering.com+1cleanroom-industries.com+1.

19.

Network requirements include:
A) Single ISP
B) Redundant high-speed links
C) WiFi only
D) No cable
Answer: B
Data centers need redundant carrier-neutral connectivity theaccessgroup.comen.wikipedia.org+9netzero-events.com+9cc-techgroup.com+9.

20.

Raised floors hinder high-density deployments when:
A) >8–10 kW/rack
B) <1 kW/rack
C) No tiles
D) No cabling
Answer: A
Underfloor cooling struggles with >8–10 kW densities en.wikipedia.org+3cleanroom-industries.com+3arxiv.org+3.

21.

Weight consideration for data center flooring?
A) No heavy load
B) Heavy tile systems: ensure structure supports raised floor
C) Floor weight is irrelevant
D) Panels stay light
Answer: B
Raised floors add load; structure must support it en.wikipedia.org+6123.net+6dataspan.com+6.

22.

Site selection should include:
A) Nearby cafes
B) Skilled technical talent

C) Movie theatres
D) Tourist zones
Answer: B
Local skilled staff availability is crucial 123.neten.wikipedia.org+2netzero-events.com+2techtarget.com+2.

23.

UFAD efficiency advantage over ceiling cooling?
A) None
B) Better air quality & lower energy use
C) More turbulence
D) Aesthetic only
Answer: B
Underfloor systems improve thermal comfort and efficiency en.wikipedia.org+1123.net+1123.net.

24.

Purpose of space reserve in planning:
A) Emergency exit
B) Capacity for future equipment
C) Office area
D) Waste area
Answer: B
Empty space allows future scalability en.wikipedia.orgen.wikipedia.org+1techtarget.com+1.

25.

Building repurposing requires evaluation of:
A) Color scheme
B) Structural capacity, floor height, power/cooling readiness
C) Landscaping
D) Interior decor
Answer: B

Retrofitting requires structural and mechanical evaluation lightboxre.com+5ibm.com+5ibm.com+5cc-techgroup.com+15mullerec.com+15ibm.com+15arxiv.org+1en.wikipedia.org+1.

26.

Presence of natural daylight contributes to:
A) Cooling
B) Staff morale and reduced lighting
C) Power supply
D) Security
Answer: B
Daylight improves work environment and energy use en.wikipedia.orgdataspan.com+1en.wikipedia.org+1.

27.

Close coupled cooling vs CRAC: benefit is:
A) Wider aisles
B) Rack-level heat removal
C) Less equipment
D) No need for raised floor
Answer: B
Close coupled cools directly at racks ibm.com+1ibm.com+1blsstrategies.com+15en.wikipedia.org+15techtarget.com+15techtarget.com.

28.

Immersion cooling downside?
A) No power savings
B) Costly retrofit and maintenance complexity
C) Cannot handle density
D) Requires raised floor
Answer: B
Retrofits are expensive and maintenance-intensive arxiv.org+1en.wikipedia.org+1ashrae.org+5en.wikipedia.org+5blsstrategies.com+5en.wikipedia.org+6arxiv.org+6en.wikipedia.org+6.

29.

Raised floor vestibules must include:
A) Lockers
B) Grommets/gaskets to seal airflow
C) Snack machine
D) Carpets
Answer: B
Seal gaps to maintain plenum efficiency [en.wikipedia.org+8theaccessgroup.com+8techtarget.com+8].

30.

Close coupled cooling chosen when rack density >:
A) 1 kW
B) 3 kW
C) 8–10 kW
D) 15 kW
Answer: C
Efficient above 8–10 kW densities [en.wikipedia.org+1theaccessgroup.com+1].

Hard (10 MCQs)

31.

PUE (Power Usage Effectiveness) benchmark best target range:
A) 2.0–3.0
B) 1.1–1.2
C) 3.0–4.0

D) 0.5–0.6
Answer: B
Top-tier PUE is ~1.1–1.2 .

32.

Raised floors generally struggle with which density using airflow?
A) <1 kW
B) 3 kW
C) 8–10 kW+
D) 0.1 kW
Answer: C
Airflow under floor insufficient above 8–10 kW[en.wikipedia.org](en.wikipedia.org).

33.

Point load vs UDL for raised floors:
A) 3 kN vs 2 kN/m^2
B) 4.5 kN vs 12 kN/m^2
C) 10 kN vs 1 kN/m^2
D) 1 kN vs 0.5 kN/m^2
Answer: B
Tiles require 4.5 kN point load and 12 kN/m^2 UDL [en.wikipedia.org+1techtarget.com+1](en.wikipedia.org).

34.

Immersion cooling compared to air improves space by:
A) 10%
B) ~67%
C) None
D) 200%
Answer: B
Space usage drops ~66% with immersion arxiv.org.

35.

Underfloor air tends to cause:
A) Perfect distribution
B) Turbulence & spotty cooling
C) No cooling
D) Floor collapse
Answer: B
Air turbulence leads to cooling spots ibm.com+1services.ku.edu+1cleanroom-industries.com.

36.

Raised floor vs low-profile: key difference is plenum?
A) Yes, raised floor uses plenum
B) No difference
C) Low-profile also requires plenum
D) Only material differs
Answer: A
Low-profile and raised floors differ by plenum provision 123.net.

37.

Data centers design for external hazards by:
A) Floor color
B) Reinforced walls and dual roofs
C) Furniture locking
D) Painting only
Answer: B
Thick walls and backup roofing mitigate disasters mullerec.com+4lightboxre.com+4theaccessgroup.com+4123.net+1cedengineering.com +1.

38.

Raised flooring maintenance best practice:
A) Never clean
B) Regular cleaning, antistatic checks
C) Only replace panels
D) Only inspect edge panels
Answer: B
Maintain cleanliness and conductivity 123.net+1en.wikipedia.org+1.

39.

Close-coupled systems allow:
A) Higher water temps and free cooling
B) No water needed
C) Higher energy use
D) No airflow
Answer: A
Higher chilled water temps enhance free cooling en.wikipedia.org+1theaccessgroup.com+1.

40.

Selecting existing building needs evaluation of:
A) Restrooms count
B) Structural load, HVAC, power biz lines
C) Staircase color
D) Lobby size
Answer: B
Existing structures require structural and MEP review .